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Methods for the manufacture of recombinant viral vectors

Abstract

The present invention relates to methods for the production of high titer recombinant viral vectors, more particularly recombinant AAV vectors, so that the methods can be effectively employed on a scale that is suitable for the practical application of gene therapy techniques.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5622856	12/1996	Natsoulis	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3101125	12/2015	EP	N/A
WO2011154520	12/2010	WO	N/A
WO2014144486	12/2013	WO	N/A

OTHER PUBLICATIONS

Kisselev L., (Structure, 2002, vol. 10: 8-9. cited by examiner
Witkowski et al., (Biochemistry 38:11643-11650, 1999. cited by examiner
Whisstock et al., (Quarterly Reviews of Biophysics 2003, vol. 36 (3): 307-340. cited by examiner
Devos et al., (Proteins: Structure, Function and Genetics, 2000, vol. 41: 98-107. cited by examiner
Alexopoulos, Annika, N., et al., “The CMV early enhancer/chicken β actin (CA6) promoter can be used to drive transgene expression during the differentiation of murine embryonic stem cells into vascular progenitors”, BMC Cell Biology, www.biomedcentral.com/1471-212/9/2. cited by applicant
Aponte-Ubillus, Juan Jose, et al., “Molecular design for recombinant adeno-associated virus (rAAV) vector production”, Applied Microbiology and Biotechnology, vol. 102, 2017, pp. 1045-1054. cited by applicant
Ayuso, E, et al., “High AAV vector purity results in serotype- and tissue-independent enhancement of transduction efficiency”, Gene Therapy, 17, 2010, pp. 503-510. cited by applicant
Cervera, Laura, et al., “Extended gene expression by medium exchange and repeated transtext transfection for recombinant protein production enhancement”, Biotechnology and Bioengineering, vol. 112: 1-13, 2015. cited by applicant
Grieger, Jostiva, C., et al., “Production of recombinant adeno-associated virus vectors using suspension HEK293 cells and continuous harvest of vector from the culture media for GMP FIX

and FLTI clinical vector", Molecular Therapy, vol. 24, pp. 287-297, 2016. cited by applicant
Grimm, Dirk, "Production methods for gene transfer vectors based on adeno-associated virus serotypes", Methods, Academic Press, vol. 28(2), pp. 146-157, 2002. cited by applicant
International Search Report for PCT/EP2020/058527 dated May 26, 2020. cited by applicant
Niwa, Hitosti, et al., "Efficient selection for high-expression a novel eukaryotic vector", Gene, vol. 108, 1991, pp. 193-200. cited by applicant
Reed, Sharon, E., et al., "Transfaction of mammalian cells using linear polyethylanimine is a simple and effective means of producing recombinant adeno-associated virus vectors", Journal of Virological Methods, 138, 2006, pp. 85-98. cited by applicant
Sharon, David, et al., "Advancements in the design and scalable production of viral gene transfer vectors", Biotechnology and Bioengineering, 115, 2018, pp. 25-40. cited by applicant
Xiao, Xiao, et al., "Production of high-titer recombinant adeno-associated virus vectors in the absence of helper adenovirus", Journal of Virology, 72:2224-2232, 1998. cited by applicant

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Background/Summary

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ELECTRONICALLY
(1) Incorporated by reference in its entirety herein is a computer-readable nucleotide/amino acid sequence listing submitted concurrently herewith and identified as follows: One 73,728 Bytes ASCII (Text) file named "SEQUENCE_LISTING.TXT," created on 16 Sep. 2021.
FIELD OF THE INVENTION
(2) The present invention relates to methods for the production of high titer recombinant viral vectors, more particularly recombinant AAV vectors, so that the methods can be effectively employed on a scale that is suitable for the practical application of gene therapy techniques.
BACKGROUND OF THE INVENTION
(3) Gene delivery is a promising method for the treatment of acquired and inherited diseases. A number of viral-based systems for gene transfer purposes have been described, including Adeno-Associated Virus (AAV)-based systems.
(4) AAVs have unique features that make them attractive as vectors for gene therapy. AAV infect a wide range of cell types. However, they are non-transforming, and are not implicated in the etiology of any human disease. Introduction of DNA to recipient host cells generally leads to long-term persistence and expression of the DNA without disturbing the normal metabolism of the cell.
(5) AAV particles are comprised of a proteinaceous capsid having three capsid proteins, VP1, VP2 and VP3, which enclose a ˜4.7 kb linear single-stranded DNA genome containing the Rep and Cap genes flanked by the viral inverted terminal repeats (ITRs). Individual particles package only one DNA molecule strand, but this may be either the plus or minus strand. Particles containing either strand are infectious, and replication occurs by conversion of the parental infecting single strand to a duplex form, and subsequent amplification, from which progeny single strands are displaced and packaged into capsids.
(6) AAV vectors can be engineered to carry a heterologous nucleotide sequence of interest (e.g., a selected gene, antisense nucleic acid molecule, ribozyme, or the like) by deleting the internal portion of the AAV genome and inserting the DNA sequence of interest between the ITRs. The

ITRs are the only sequences required in cis for replication and packaging of the vector genome containing the heterologous nucleotide sequence of interest. The heterologous nucleotide sequence is also typically linked to a promoter sequence capable of driving gene expression in the patient's target cells under certain conditions. Termination signals, such as polyadenylation sites, are usually included in the vector.
(7) The rep and cap AAV gene products provide functions for replication and encapsidation of the vector genome, respectively, and it is sufficient for them to be present in trans.
(8) Despite the potential benefits of gene therapy as a treatment for human genetic diseases, serious practical limitations stand in the way of its widespread use in the clinic. In this sense, it is necessary to produce big amounts of rAAV particles in order to produce clinically effective doses. Production of a large number of particles using current technology requires a large number of producer cells. At a laboratory scale, it would require thousands of tissue culture flasks. At a commercial scale, more efficient and intensive production platforms are needed to reach clinical application. The benefits of improving both, the number of producing cells and the particle yield per cell, will be very significant from a commercial production standpoint.
(9) Accordingly, in the development of recombinant AAV vectors such as those for use in gene therapy, there is a need for strategies that achieve a high titer of AAV so that the methods can be effectively employed on a scale that is suitable for the practical application of gene therapy techniques.
(10) The present disclosure provides methods for achieving these competing goals and demonstrates that such techniques can be employed for the large-scale production of recombinant AAV vector preparations.
SUMMARY OF THE INVENTION
(11) The inventors of the present invention have found that, surprisingly, recombinant viral production, more particularly rAAV production, can be significantly improved by performing repeated transfections. As it is shown in the Examples below, a sustained level of gene expression over time is obtained when repeated rounds of transfection are performed, as opposed to the conventional approach which entails a single transfection round.
(12) Thus, in a first aspect, the invention refers to a method for the production of a recombinant viral vector said method comprising the steps of: a) co-transfecting a suitable cell culture with at least two plasmid vectors, said plasmid vectors comprising a heterologous nucleotide sequence and replication and packaging gene sequences; b) culturing said cells under conditions allowing viral replication and packaging; c) recovering the viral vectors produced in step b) and retaining the cells in the cell culture under conditions allowing further division and growth; d) re-transfecting the cells according to step c) with the plasmid vectors according to step a); and e) repeating steps b) to c).
(13) Furthermore, the inventors have found that the combination of optimized plasmids for the transfection, results in higher yield of rAAV than production with the standard plasmids commonly used for AAV vector production. Moreover, as it is shown in the Examples below, the inventors have surprisingly found that the use of the optimized plasmids according to the invention results in lower reverse packaging of bacterial sequences.
(14) It is thus a further aspect of the present invention a method for h production of a recombinant AAV said method comprising the steps of: a) co-transfecting a suitable cell with i) a first plasmid vector comprising a heterologous nucleoli sequence flanked by ITRs; ii) a second plasmid vector comprising from 5′ to 3′ an AAV rep coding region, an AAV cap coding region and a nucleotide sequence comprising an AAV p5 promoter region, and iii) a third plasmid vector comprising adenovirus helper functions including VA-RNA, E2A and E4 sequences, wherein said plasmid does not contain E3, pTB (E2B), Ad ITR and protease sequences; b) culturing said cell under conditions allowing AAV replication and packaging; and c) recovering the AAVs produced in step b).
(15) The inventors have surprisingly found that by using a combination of optimized plasmids in the transfection, reverse packaging is greatly reduced and higher vector genome yield is obtained.

(16) Thus, in another aspect, the invention refers to a plasmid vector comprising: a) a heterologous nucleotide sequence flanked by ITRs; and b) a stuffer DNA sequence located outside said ITRs and adjacent to one ITR, wherein said stuffer sequence has a length between 4400 Kb and 4800 Kb so that the plasmid backbone size is above 5 Kb, preferably between 7000 bp and 7500 bp; wherein said plasmid vector does not contain a F1Ori nucleotide sequence in the backbone sequence.
(17) In a further aspect, the invention refers to a plasmid vector comprising adenovirus helper functions including VA-RNA, E2A and E4 sequences, wherein said plasmid does not contain E3, pTB(E2B), and Ad ITR protease sequences.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**. Yields of AAV9-CAG-cohSgsh expressed in total vg produced by triple transfection with a set of standard plasmids or a set of optimized plasmids.
(2) FIG. **2**. Copies of bacterial sequence present in AAV9-CAG-cohSgsh produced by triple transfection with a set of standard plasmids or a set of optimized plasmids,
(3) FIG. **3**. Effect of the optimized plasmid pcohSgsh-900 alone or in combination with the other optimized helper plasmids (pRepCap9-809 and pAdhelper861) on the AAV vector yield (total vg).
(4) FIG. **4**. Reverse packaging of bacterial sequences. Effect of the optimized oversized plasmid pcohSgsh-900 alone or in combination with the other optimized helper plasmids (pRepCap9-809 and pAdhelper861).
(5) FIG. **5**. Reverse packaging of bacterial sequences. Effect of the optimized oversized plasmid pcohSgsh-900 when combined with the other optimized helper plasmids (pRepCap9-809 and pAdhelper861),
(6) FIG. **6**. Total vg recovered after transient transfection with pohIDS-874 plasmid. Cell lysis and total vg recovered in the culture supernatant after performing several re-transfection rounds. Time is represented in hours post transfection (hpt).
(7) FIG. **7**, Total vg recovered after transient transfection with pohSGSH-900 plasmid, Cell lysis and total vg recovered in the culture supernatant after performing several re-transfection rounds. Time is represented in hours post transfection (hpt).
DEPOSIT OF MICROORGANISMS
(8) The plasmid pAdHelper861 was deposited on Dec. 5, 2018, under access number DSM 32965 at the DSMZ—Deutsche Sammlung von Mikroorganismen und Zeilkulturen, Inhoffenstraße 7 B, D-38124 Braunschweig, Federal Republic of Germany.
(9) The plasmid pcohSgsh-827 was deposited on Dec. 5, 2018, under access number DSM 32966 at the DSMZ—Deutsche Sammlung von Mikroorganismen and Zellkulturen, Inhoffenstraße 7 B, D-38124 Braunschweig, Federal Republic of Germany.
(10) The plasmid pcohSgsh-900 was deposited on Dec. 5, 2018, under access number DSM 32967 at the DSMZ—Deutsche Sammlung von Mikroorganismen und Zellkulturen, Inhoffenstraße 7 B, D-38124 Braunschweig, Federal Republic of Germany.
(11) The plasmid pohIDS-874 was deposited on Dec. 5, 2018, under access number DSM 32968 at the DSMZ—Deutsche Sammlung von Mikroorganismen und Zellkulturen, Inhoffenstraße 7 B, D-38124 Braunschweig, Federal Republic of Germany.
(12) The plasmid pRepCap9-809 was deposited on Dec. 5, 2018, under access number DSM 32969 at the DSMZ—Deutsche Sammlung von Mikroorganismen und Zellkulturen, Inhoffenstraße 7 B, D-38124 Braunschweig, Federal Republic of Germany.
Definitions
(13) A “vector” as used herein refers to a macromolecule or association of macromolecules that

comprises or associates with a polynucleotide and which can be used to mediate delivery of the polynucleotide to a cell. Illustrative vectors include, for example, plasmids, viral vectors, liposomes and other gene delivery vehicles.

(14) The terms "adeno-associated virus", "AAV virus", "AAV virion," "AAV viral particle" and "AAV particle", used as synonyms herein, refer to a viral particle composed of at least one capsid protein of AAV (preferably composed of all capsid proteins of a particular AAV serotype) and an encapsulated polynucleotide corresponding to the AAV genome. The wild-type AAV refers to a virus that belongs to the genus Dependovirus, family Parvoviridae. The wild-type AAV genome is approximately 4.7 Kb in length and consists of a single stranded deoxyribonucleic acid (ssDNA) that can be positive or negative-sensed. The wild-type genome includes inverted terminal repeats (ITR) at both ends of the DNA strand, and three open reading frames (ORFs). The ORF rep encodes for four Rep proteins necessary for AAV lifecycle. The ORF cap contains nucleotide sequences encoding capsid proteins: VP1, VP2 and VP3, which interact to form a capsid of icosahedral symmetry. Finally, the AAP ORF, which overlaps with the Cap ORF, encodes for the AAP protein that appears to promote capsid assembly. If the particle comprises a heterologous polynucleotide (i.e. a polynucleotide different from a wild-type AAV genome, such as a transgene to be delivered to a mammalian cell) flanked by AAV ITRs, then it is typically known as "AAV vector particle" or "AAV viral vector" or "AAV vector" or "recombinant AAV vectors". The invention also encompasses the use of double stranded AAV or self-complimentary AAV, also called dsAAV or scAAV.

(15) The term "adeno-associated virus ITRs" or "AAV ITRs", as used herein, refers to the inverted terminal repeats present at both ends of the DNA strand of the genome of an AAV. The ITR sequences are required for efficient multiplication of the AAV genome. Another property of these sequences is their ability to form a hairpin. This characteristic contributes to their self-priming, which allows the primase-independent synthesis of the second DNA strand. The ITRs have also been shown to be required for both integration of the wild-type AAV DNA into the host cell genome (e.g. in the human 19.sup.th chromosome for serotype 2 MV) and rescue from it, as well as for efficient encapsidation of the MV DNA into a fully assembled, deoxyribonuclease-resistant AAV particle. The ITR sequences are about 145 bp in length. Preferably, the entire sequences of the ITRs are used in the genome of the AAV viral vector, although some degree of minor modification of these sequences is permissible. A wild-type ITR sequence may be altered by insertion, deletion or truncation, as long as the ITR mediates the desired functions, e.g. replication, nicking, virus packaging, integration, and/or provirus rescue. Procedures for modifying these ITR sequences are well known in the art. The ITR may be from any wild-type AAV, including but not limited to serotypes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12 or any other AAV known or later discovered. The AAV comprises two ITRs, which may be the same or different. Further, the two AAV ITRs can be from the same AAV serotype as the AAV capsid, or can be different. In a preferred embodiment, the 5′ and 3′ AAV ITRs derive from AAV1, AAV2, AAV4, AAV5, AAV7, AAV8 and/or AAV9. Preferably ITRs are from AAV2, AAV8 and/or AAV9 being AAV2 the most preferred. In one embodiment, the AAV2 ITRs are selected to generate a pseudotyped AAV (i.e. an AAV having capsid and ITRs derived from different serotypes).

(16) The expression "recombinant viral genome", as used herein, refers to an MV genome in which at least one extraneous polynucleotide is inserted into the naturally occurring AAV genome. The genome of the AAV according to the invention typically comprises the cis-acting 5′ and 3′ inverted terminal repeat sequences (ITRs) and an expression cassette. A "rAAV vector" as used herein refers to an AAV vector comprising a polynucleotide sequence not of AAV origin (i.e., a polynucleotide heterologous to AAV), typically a sequence of interest for the genetic transformation of a cell. In preferred vector constructs of this invention, the heterologous polynucleotide is flanked by two AAV inverted terminal repeat sequences (ITRs).

(17) An "AAV virus" or "AAV viral particle" refers to a viral particle composed of at least one

AAV capsid protein (preferably by all of the capsid proteins of a wild-type AAV) and an encapsidated polynucleotide. If the particle comprises a heterologous polynucleotide (i.e. a polynucleotide other than a wild-type AAV genome such as a transgene to be delivered to a mammalian cell), it is typically referred to as a “rAAV vector particle” or simply a “rAAV vector”.
(18) “Packaging” refers to a series of intracellular events that result in the assembly of the capsid proteins and encapsidation of the vector genome to form an AAV particle.
(19) AAV “rep” and “cap” genes refer to polynucleotide sequences encoding replication and encapsidation proteins of adeno-associated virus. They have been found in all AAV serotypes examined, and are described below and in the art. AAV rep and cap are referred to herein as AAV “packaging genes”.
(20) The term “GAG promoter” refers to the combination formed by the cytomegalovirus early enhancer element, chicken β-actin promoter and 3′ splice sequence derived from the rabbit beta-globin gene (See Alexopoulou A, et al., BMC Cell Biology 2008; 9(2): 1-11, Niwa et al, Gene. 1991 Dec. 15; 108(2):193-9).
(21) The term “polynucleotide” refers to a polymeric form of nucleotides of any length, including deoxyribonucleotides or ribonucleotides, or analogs thereof. A polynucleotide may comprise modified nucleotides, such as methylated nucleotides and nucleotide analogs, and may be interrupted by non-nucleotide components. If present, modifications to the nucleotide structure may be imparted before or after assembly of the polymer. The term polynucleotide, as used herein, refers interchangeably to double- and single-stranded molecules. Unless otherwise specified or required, any embodiment of the invention described herein that is a polynucleotide encompasses both the double-stranded form and each of two complementary single-stranded forms known or predicted to make up the double-stranded form.
(22) A “gene” refers to a polynucleotide containing at least one open reading frame that is capable of encoding a particular protein after being transcribed and translated, or polynucleotide containing at least one non-coding RNA.
(23) The term “nucleotide sequence” refers to a nucleic acid molecule, either DNA or RNA, containing deoxyribonucleotides or ribonucleotides. The nucleic acid may be double stranded, single stranded, or contain portions of both double stranded or single stranded sequence.
(24) The term “codify” refers to the genetic code that determines how a nucleotide sequence is translated into a polypeptide, or a protein. The order of the nucleotides in a sequence determines the order of amino acids along a polypeptide or a protein.
(25) “Recombinant”, as applied to a polynucleotide, means that the polynucleotide is the product of various combinations of cloning, restriction or ligation steps, and other procedures that result in a construct that is distinct from a polynucleotide found in nature. A recombinant virus is a viral particle comprising a recombinant polynucleotide. The terms respectively include replicates of the original polynucleotide construct and progeny of the original virus construct.
(26) A “control element” or “control sequence” is a nucleotide sequence involved in an interaction of molecules that contributes to the functional regulation of a polynucleotide, including replication, duplication, transcription, splicing, translation, or degradation of the polynucleotide. The regulation may affect the frequency, speed, or specificity of the process, and may be enhancing or inhibitory in nature. Control elements known in the art include, for example, transcriptional regulatory sequences such as promoters and enhancers. A promoter is a DNA region capable under certain conditions of binding RNA polymerase and initiating transcription of a coding region usually located downstream (in the 3′ direction) from the promoter.
(27) “Operatively linked” or “operably linked” refers to a juxtaposition of genetic elements, wherein the elements are in a relationship permitting them to operate in the expected manner. For instance, a promoter is operatively linked to a coding region if the promoter helps to initiate transcription of the coding sequence. There may be intervening residues between the promoter and coding region so long as this functional relationship is maintained.

(28) An “expression vector” is a vector comprising a region, which encodes a polypeptide of interest, and is used for effecting the expression of the protein in an intended target cell. An expression vector also comprises control elements operatively linked to the encoding region to facilitate expression of the protein in the target. The combination of control elements and a gene or genes to which they are operably linked for expression is sometimes referred to as an “expression cassette,” a large number of which are known and available in the art or can be readily constructed from components that are available in the art.
(29) “Heterologous” means derived from a genotypically distinct entity from that of the rest of the entity to which it is being compared. For example, a polynucleotide introduced by genetic engineering techniques into a plasmid or vector derived from a different species is a heterologous polynucleotide. A promoter removed from its native coding sequence and operatively linked to a coding sequence with which it is not naturally found linked is a heterologous promoter. In particular, the term “heterologous nucleotide sequence” as used herein includes coding as well as non-coding nucleotide sequences.
(30) “Genetic alteration” refers to a process wherein a genetic element is introduced into a cell other than by mitosis or meiosis. The element may be heterologous to the cell, or it may be an additional copy or improved version of an element already present in the cell. Genetic alteration may be effected, for example, by transfecting a cell with a recombinant plasmid or other polynucleotide through any process known in the art, such as electroporation, calcium phosphate precipitation, or contacting with a polynucleotide-liposome complex. Genetic alteration may also be effected, for example, by transduction or infection with a DNA or RNA virus or viral vector. Preferably, the genetic element is introduced into a chromosome or mini-chromosome in the cell; but any alteration that changes the phenotype and/or genotype of the cell and its progeny is included in this term.
(31) The terms “polypeptide”, “peptide” and “protein” are used interchangeably herein to refer to polymers of amino acids of any length. The terms also encompass an amino acid polymer that has been modified; for example, disulfide bond formation, glycosylation, lipidation, or conjugation with a labeling component.
(32) An “isolated” plasmid, virus, or other substance refers to a preparation of the substance devoid of at least some of the other components that may also be present where the substance or a similar substance naturally occurs or is initially prepared from. Thus, for example, an isolated substance may be prepared by using a purification technique to enrich it from a source mixture. Enrichment can be measured on an absolute basis, such as weight per volume of solution, or it can be measured in relation to a second, potentially interfering substance present in the source mixture. Increasing enrichments of the embodiments of this invention are increasingly more preferred. Thus, for example, a 2-fold enrichment is preferred, 10-fold enrichment is more preferred, 100-fold enrichment is more preferred, 1000-fold enrichment is even more preferred.
(33) An “individual” or “subject” treated in accordance with this invention refers to vertebrates, particularly members of a mammalian species, and includes but is not limited to domestic animals, sports animals, and primates, including humans.
(34) “Treatment” of an individual or a cell is any type of intervention in an attempt to alter the natural course of the individual or cell at the time the treatment is initiated. For example, treatment of an individual may be undertaken to decrease or limit the pathology caused by any pathological condition, including (but not limited to) an inherited or induced genetic deficiency, infection by a viral, bacterial, or parasitic organism, a neoplastic or aplastic condition, or an immune system dysfunction such as autoimmunity or immunosuppression. Treatment includes (but is not limited to) administration of a composition, such as a pharmaceutical composition, and administration of compatible cells that have been treated with a composition. Treatment may be performed either prophylactically or therapeutically; that is, either prior or subsequent to the initiation of a pathologic event or contact with an etiologic agent.

(35) The term “effective amount” refers to an amount of a substance sufficient to achieve the intended purpose. For example, an effective amount of an expression vector to increase sulfamidase activity is an amount sufficient to reduce glycosaminoglycan accumulation. A “therapeutically effective amount” of an expression vector to treat a disease or disorder is an amount of the expression vector sufficient to reduce or remove the symptoms of the disease or disorder. The effective amount of a given substance will vary with factors such as the nature of the substance, the route of administration, the size and species of the animal to receive the substance and the purpose of giving the substance. The effective amount in each individual case may be determined empirically by a skilled artisan according to established methods in the art.
(36) The term “gene therapy” refers to the transfer of genetic material (e.g. DNA or RNA) of interest into a host to treat or prevent a genetic or acquired disease or condition. The genetic material of interest encodes a product (e.g. a protein polypeptide, peptide or functional RNA) whose production in vivo is desired. For example, the genetic material of interest can encode an enzyme, hormone, receptor, or polypeptide of therapeutic value.

DETAILED DESCRIPTION OF THE INVENTION

(37) It is an object of this invention to provide methods for generating high titer preparations of recombinant viral vectors, in particular, recombinant adeno-associated viral vectors (rAAV) so that the methods can be effectively employed on a scale that is suitable for the practical application of gene therapy procedures.
(38) As it is shown in the Examples accompanying the present invention, the inventors have found that, surprisingly, rAAV production can be significantly improved by performing repeated rounds of transfection, i.e. by re-transfections, of the cell culture with the plasmid vectors used for triple or double transfection.
(39) Thus, in a first aspect, the invention refers to a method for the production of a recombinant viral vector said method comprising the steps of: a) co-transfecting a suitable cell culture with at least two plasmid vectors, said plasmid vectors comprising a heterologous nucleotide sequence and replication and packaging gene sequences; b) culturing said cells under conditions Rowing viral replication and packaging; c) recovering the viral vectors produced in step b) and retaining the cells in the cell culture under conditions allowing further division and growth; d) re-transfecting the cells according to step c) with the plasmid vectors according to step a); and e) repeating steps b) to c).
(40) In a particular embodiment of the invention, said recombinant viral vector is selected from the group consisting of an adenovirus, adeno-associated virus (AAV), alphavirus, flavivirus, herpes simplex virus (HSV), measles virus, rhabdovirus, retrovirus, lentivirus, Newcastle disease virus (NDV), poxvirus, and picornavirus. In a particular embodiment, said viral vector is selected a retroviral vector, an adenoviral vector or an AAV vector.
(41) In a preferred embodiment, said viral vector is an AAV vector.
(42) AAV according to the present invention include any serotype of the AAV known serotypes. In general, the different serotypes of AAV have genomic sequences with a significant homology, providing an identical series of genetic functions, produce virions that are essentially equivalent in physical and functional terms, and replicate and assemble through practically identical mechanisms. In particular, the AAV of the present invention may belong to the serotype 1 of AAV (AAV1), AAV2, AAV3 (including types 3A and 3B), AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, avian AAV, bovine AAV, canine AAV, equine AAV, ovine AAV, and any other AAV. Examples of the sequences of the genome of the different AAV serotypes may be found in the literature or in public databases such as GenBank. See GenBank accession numbers AF028704.1 (AAV6), NC006260 (AAV7), NC006261 (AAV8), and AX753250.1 (AAV9). In a preferred embodiment, the AAV vector of the invention is of a serotype selected from the group consisting of the AAV2, AAV5, AAV7, AAV8, AAV9, AAV10 and AAVrh10 serotypes. In a preferred embodiment, said AAV vector of the invention is of serotype 9.
(43) In a particular embodiment, the invention refers to a method for the production of a

recombinant AAV said method comprising the steps of: a) co-transfecting a suitable cell culture with at least two plasmid vectors said vectors comprising a heterologous nucleotide sequence flanked by ITRs, AAV rep and AAV cap gene sequences, and adenovirus helper functions sequences; b) culturing said cells under conditions allowing AAV replication and packaging; c) recovering the AAVs produced in step b) and retaining the cells in the cell culture under conditions allowing further division and growth, d) re-transfecting the cells according to step c) with the plasmid vectors according to step a), and e) repeating steps b) to c).
(44) As it is described in the art, genetic material can be introduced into cells using any of a variety of means to transform or transduce such cells. By way of illustration, such techniques include, for example, transfection with bacterial plasmids. Indeed, the plasmid vectors according to step a) of the method of the invention can be introduced into a cell using a variety of transfection techniques. Such transfection methods have been described in the art and include, for example, calcium phosphate co-precipitation, direct micro-injection into cultured cells, electroporation, liposome mediated gene transfer, lipid-mediated transfection or nucleic acid delivery using high-velocity microprojectiles. Other suitable transfection media include strontium phosphate, polycationic polymers, e.g., Superfect (QIAGEN™) liposomes, and cationic polymers such as polyethylenimine (PEI). In a preferred embodiment, the plasmid vectors according to the method of the invention are transfected using PEI. In this case, PEI/DNA complexes are formed by adding PE to plasmid DNA prior to its addition to the cell culture.
(45) Any of these techniques can be used to introduce one or more, exogenous DNA moieties, such as vector constructs, into suitable host cells. Generally, the exogenous DNA must traverse the host cell plasma membrane in order to be exposed to the cell's transcription and replication machinery.
(46) The resulting cell can be transiently transfected with the exogenous nucleic acid molecule, i.e., the exogenous DNA will not be integrated into the genome of a transfected cell, but rather will exist episomally. Alternatively, the resulting cell can be stably transfected, i.e., the nucleic acid molecule will become covalently linked with the host cell genome or will be maintained and replicated as an episomal unit which can be passed on to progeny cells (e.g., capable of extra-chromosomal replication at a sufficient rate).
(47) According to the method of the invention, the plasmid vectors to be transfected comprise a heterologous nucleotide sequence flanked by the ITRs. AAV rep and AAV cap gene sequences, and adenovirus helper functions sequences.
(48) The heterologous nucleotide sequence or polynucleotide is typically of interest because of a capacity to provide a function to a target cell in the context of gene therapy, such as up- or down-regulation of the expression of a certain phenotype. Such a heterologous polynucleotide or “transgene” will generally be of sufficient length to provide the desired function or encoding sequence.
(49) Where transcription of the heterologous polynucleotide is desired in the intended target cell, it can be operably linked to its own or to a heterologous promoter, depending for example on the desired level and/or specificity of transcription within the target cell, as is known in the art. Various types of promoters and enhancers are suitable for use in this context. Constitutive promoters provide an ongoing level of gene transcription, and are preferred when it is desired that the therapeutic polynucleotide be expressed on an ongoing basis. Inducible promoters generally exhibit low activity in the absence of the inducer, and are up-regulated in the presence of the inducer. They may be preferred when expression is desired only at certain times or at certain locations, or when it is desirable to titrate the level of expression using an inducing agent. Promoters and enhancers may also be tissue-specific, that is, they exhibit their activity only in certain cell types, presumably due to gene regulatory elements found uniquely in those cells.
(50) Illustrative examples of promoters are the SV40 late promoter from simian virus 40, the Baculovirus polyhedron enhancer/promoter element, Herpes Simplex Virus thymidine kinase (HSV tk), the immediate early promoter from cytomegalovirus (CMV), the CMV early enhancer/chicken

β actin (CAG) promoter and various retroviral promoters including LTR elements. Inducible promoters include heavy metal ion inducible promoters (such as the mouse mammary tumor virus (mMTV) promoter or various growth hormone promoters), and the promoters from T7 phage which are active in the presence of T7 RNA polymerase. By way of illustration, examples of tissue-specific promoters include various surfactant proteins promoters (for expression in the lung), myosin promoters (for expression in muscle), and albumin promoters or human α1-antitrypsin hAAT (for expression in the liver). A large variety of other promoters are known and generally available in the art, and the sequences for many such promoters are available in sequence databases such as the GenBank database. In a preferred embodiment, the CAG promoter is used.
(51) Where translation is also desired in the intended target cell, the heterologous polynucleotide will preferably also comprise control elements that facilitate translation (such as a ribosome binding site or “RBS” and/or a polyadenylation signal). Accordingly, the heterologous polynucleotide will generally comprise at least one coding region operatively linked to a suitable promoter, and may also comprise, for example, an operatively linked enhancer, ribosome binding site and/or poly-A signal. The heterologous polynucleotide may comprise one coding region, or more than one coding regions under the control of the same or different promoters. The entire unit, containing a combination of control elements and coding or non-coding region, is often referred to as an expression cassette. In a particular embodiment, the heterologous polynucleotide according to the method of the invention contains a poly-A signal.
(52) The heterologous polynucleotide is integrated by recombinant techniques into or preferably in place of the AAV genomic coding region and is generally flanked on either side by AAV inverted terminal repeat (ITR) regions. Alternatively, vector constructs with only one ITR can be employed. In a particular embodiment, said heterologous nucleotide sequence is flanked by two ITRs.
(53) Given the encapsidation size limit of the AAV particles, insertion of a large heterologous polynucleotide into the genome necessitates removal of a portion of the AAV sequence. Removal of one or more AAV genes is in any case desirable, to reduce the likelihood of generating replication-competent AAV (“RCA”). Accordingly, encoding or promoter sequences for rep, cap, or both, are preferably removed, since the functions provided by these genes can be provided in trans.
(54) In one embodiment, the heterologous nucleotide sequence is flanked by AAV ITRs, and the AAV packaging genes to be provided in trans, are introduced into the host cell in separate vector plasmids.
(55) The rep gene is expressed from two promoters, p5 and p19, and produces four proteins designated Rep78, Rep68, Rep52 and Rep40. Only Rep78 and Rep68 are required for AAV duplex DNA replication, but Rep52 and Rep40 appear to be needed for progeny, single-strand DNA accumulation. Rep68 and Rep78 bind specifically to the hairpin conformation of the AAV ITR and possess several enzyme activities required for resolving replication at the AAV termini. Rep78 and Rep68, also exhibit pleiotropic regulatory activities including positive and negative regulation of AAV genes and expression from some heterologous promoters, as well as inhibitory effects on cell growth.
(56) The cap gene encodes capsid proteins VP1, VP2, and VP3, These proteins share a common overlapping sequence, but VP1 and VP2 contain additional amino terminal sequences transcribed from the p40 promoter by use of alternate initiation codons. All three proteins are required for effective capsid production.
(57) Packaging of an AAV vector into viral particles still relies on the presence of a suitable helper virus for AAV or the provision of helper virus functions. According to the method of the present invention, helper virus function gene sequences are provided in plasmid vectors. The presence of significant quantities of infectious helper virus in a preparation of AAV vectors is problematic in that the preparation is intended for use in human administration.
(58) In a particular embodiment of the method of the invention, transfection is performed using two plasmid vectors wherein one of said plasmid vectors comprises a heterologous nucleotide sequence

flanked by ITRs and the second plasmid vector comprises the AAV rep and AAV cap gene sequences and adenovirus helper functions sequences. In a preferred embodiment, said second plasmid vector comprises from 5′ to 3′ an AAV rep coding region, an AAV cap coding region and a nucleotide sequence comprising an AAV p5 promoter region.

(59) In a preferred embodiment of the method of the invention, transfection is performed using three plasmid vectors, i.e. the host cells are triple-transfected with at least one vector encoding a heterologous nucleotide sequence of interest, at least one vector encoding AAV rep and cap genes, and at least one vector encoding, adenoviral accessory functions.

(60) Selection of suitably altered cells may be conducted by any technique in the art. For example, the polynucleotide sequences used to alter the cell may be introduced simultaneously with or operably linked to one or more detectable or selectable markers as is known in the art. By way of illustration, one can employ a drug resistance gene as a selectable marker. Drug resistant cells can then be picked and grown, and then tested for expression of the desired sequence—i.e., a packaging gene product, or a product of the heterologous polynucleotide, as appropriate. Testing for acquisition, localization and/or maintenance of an introduced polynucleotide can be performed using DNA hybridization-based techniques (such as Southern blotting and other procedures as known in the art). Testing for expression can be readily performed by Northern analysis of RNA extracted from the genetically altered cells, or by indirect immunofluorescence for the corresponding gene product.

(61) According to step b) of the method of the invention, the transfected cells are thus cultured under conditions allowing viral vector replication and packaging. In a particular embodiment, the cells are cultured under conditions allowing rAAV replication and packaging.

(62) Several criteria influence selection of cells for use in producing rAAV particles as described herein. The more preferred cells and cell lines are those that can be easily grown in culture so as to facilitate large-scale production of recombinant AAV vector preparations. Where large-scale production is desired, the choice of production method will also influence the selection of the host cell. For example, some production techniques and culture vessels or chambers are designed for growth of adherent or attached cells, whereas others are designed for growth of cells in suspension. In the latter case, the host cell would thus preferably be adapted or adaptable to growth in suspension. According to the present invention, large-scale production of rAAVs is desired.

(63) A variety of cells lines are contemplated for use in the large-scale production of rAAV. Particularly suitable for cell culture of rAAV are Human Embryo Kidney (HEK) 293 cell lines, either adherent or selected for growth in suspension. Examples of other suitable cell lines to produce rAAV include: Vero cells, HeLa cells, and CHO cell lines. As used herein, the term “cell line” refers to a population of cells capable of continuous or prolonged growth and division in vitro. Often, cell lines are clonal populations derived from a single progenitor cell. It is known in the art that spontaneous or induced changes can occur in karyotype during storage or transfer of such clonal populations. Therefore, cells derived from the cell line referred to may not be precisely identical to the ancestral cells or cultures Nevertheless, the term “cell line” includes such variants.

(64) Once a cell line is selected, the cell culture device of choice is seeded with cells at a density suitable to support cell culture. The density of cells used to seed a particular device will depend on the size of the device. A variety of volumes may be used to grow the cells in the culture device of choice. The volume of medium used will vary according to the size of the culture device used to grow the culture cells. Large-scale production methods such as suspension culture may be used. AAV particles are then collected, and isolated from the cells used to prepare them.

(65) Cells are cultured under conditions that are permissive for the replication AAV and packaging of the rAAV vector. Culture time is preferably adjusted to correspond to peak production levels. Preferably, at least 100 viral particles are produced per cell: more preferably at least about 1000 per cell, still more preferably at least about 10,000 per cell. Preferably, at least 0.5×10.sup.6, more preferably at least about 1×10.sup.6, even more preferably at least about 2×10.sup.6RU/ml AAV

vectors are produced per 2×10.sup.5 cells during the culture period.

(66) Various growth media may be used in the disclosed invention to achieve large-scale cell growth and AAV production. The selection of growth medium varies depending on the type of cells being cultured. In a particular embodiment, said cells are mammalian cells. In a preferred embodiment, said cell is Human Embryo Kidney (HEK) 293 cell lines. More preferably, said cells are cells suitable for being grown in suspension.

(67) For the culture of adherent cells, roller bottles (cylindrical tissue culture flasks rotated at a given velocity) are used in order to provide the necessary surface for cell attachment. Alternatively, micro-carriers are also a suitable system to support adherent cell cultures. For the culture of suspension cells, stirred tank bioreactors are the most indicated systems, since they enable to reach high cell density cultures that in turn can provide large amounts of rAAV after purification.

(68) In a particular embodiment of the invention, a rocking-motion-type bioreactor is used. In another particular embodiment a stirred tank bioreactor could also be used. Once the cell culture in the reactor reaches a given point, transfection of cells is performed as mentioned above.

(69) According to the method of the invention, step c) comprises recovering the AAVs produced in step b) and maintaining the cells in the cell culture under conditions allowing further division and growth. In a preferred embodiment of the method, step b) is performed culturing said cell in suspension in agitated liquid medium.

(70) According to a particular embodiment of the method of the invention, in step b) the AAVs produced are secreted to the supernatant of the cell culture. Thus, according to step c) of the method of the invention, the rAAVs are harvested from the supernatant and the cells are kept in the cell culture for further division and growth. The spent media or supernatant is then collected together with the rAAVs so produced since, as it is shown in the Examples below, AAVs are secreted to the supernatant of the cell culture without the need of cell lysis.

(71) In a particular embodiment, the cell media of the cell culture is exchanged before re-transfection is performed, i.e. before step d). In a particular embodiment, the cell media exchange is performed by centrifugation. In a preferred embodiment, media exchange is performed by perfusion. In a more preferred embodiment, continuous media exchange is performed by perfusion. More preferably, said culture medium is automatically exchanged by a perfusion system. By this system, the culture is replenished with fresh medium while cell-free supernatant is removed using a cell retention device. This process is preferably performed at a constant harvest flow rat. In this regard, in a preferred embodiment, new, fresh, culture media is added to the retained cells in order to allow them to further divide and grow. The constant addition of nutrients and removal of toxic metabolites allows perfusion cultures to reach and sustain high cell densities over many weeks.

(72) The inventors have found that surprisingly, it is possible to re-transfect the retained cells in the cell culture with the plasmid vectors as described above for step a) so that the production of AAVs can be prolonged. In this sense, more rAAVs are produced using the cells retained in the cell culture device which are allowed to further divide and grow. Thus, the inventors have found that by performing repeated rounds of transfection, production of rAAVs can be extended over time using the same cell culture. Moreover, as it can be seen in the Examples accompanying the present invention, the inventors have found that by doing these medium exchanges, AAVs are secreted to the supernatant of the cell culture without the need of cell lysis and can be harvested every time the medium exchange is performed before each re-transfection round. Another advantage of the proposed methodology is that rAAVs are recovered from the supernatant, this being an advantage from the purification point of view as cells do not need to be lysed to harvest the AAVs and thus, less contamination with host cell DNA and host cell protein occurs.

(73) Hence, the method of the invention comprises a step d) of re-transfecting the retained cells according to step c) with the plasmid vectors according to step a).

(74) According to step d) of the method of the invention, re-transfection or repeated rounds of transfection are performed in the cell culture, as opposed to the conventional transient gene

expression (TGE) approach which entails a single transfection round without medium exchange.
(75) Moreover, the inventors have found that surprisingly said re-transfection and recovering steps can be repeated more than two times using the same cell culture, in a particular embodiment of the method of the invention, steps d), b) and c) are repeated at least one more time after recovering step c). In another particular embodiment steps d), b) and c) are repeated at least twice after recovering step c). In a more particular embodiment, steps d), b) and c) are repeated at least three times after recovering step c). In a preferred embodiment, steps d), b) and c) are repeated at least four times after recovering step c). In a more particular embodiment, steps d), b) and c) are repeated between one and three times after recovering step c). In a more particular embodiment, steps d), b) and c) are repeated between one and two times after recovering step c). In a preferred embodiment, the method of the invention includes one repetition of steps d), b) and c) after recovering step c). In a preferred embodiment of the method of the invention, steps d), b) and c) are repeated one more time after recovering step c).
(76) As mentioned before, in a particular embodiment of the method of the invention, continuous media exchange is performed by perfusion. In a preferred embodiment, every time transfection is to be performed, perfusion stands. This allows plasmid vectors to enter the cell. In a more preferred embodiment, perfusion stands for a period of time of at least 1.5 hours, more preferably, of at least 2 hours, more preferably of at least 3 hours, even more preferably of at least 4 hours. In a preferred embodiment, there is a minimum interval of time between each re-transfection round. In a more preferred embodiment, said re-transfections are performed after an interval of time of at least 36 hours, more preferably, of at least 48 hours.
(77) In another particular embodiment, said method comprises an additional step of lysing the cells in the supernatant after all rounds of re-transfection and recovering are finished.
(78) There are several well-known techniques in the state of the art that can be used for cell disruption or cell lysis. Although freeze-thawing and/or sonication can be used to disrupt the cells, such techniques are not very suitable to large-scale preparations. Mechanical lysis techniques are thus preferable in those regards. Detergents and other chemical agents can also be employed to mediate or facilitate lysis. Treatment of lysates with nucleases (such as benzonase) has been found to be helpful for reducing viscosity and improving filterability. Clarification, e.g. by microfiltration to separate vector from at least some portion of the cellular debris, is also helpful for promoting recovery and purification. In a particular embodiment of the invention, said lysing step according to the method of the invention comprises using a nuclease, more preferably, said nuclease is benzonase.
(79) In a particular embodiment of the method of the invention, the supernatant collected containing the rAAVs of the invention as described above is further processed so that the rAVVs are purified. In this regard, in another particular embodiment, the process is performed in a bioreactor coupled to a cell retention membrane, so that the cells are retained inside the reactor system while the AAVs are collected via the membrane, in a clean supernatant, free of cellular debris and therefore making purification much easier.
(80) In order to be particularly useful for the production of AAV for gene therapy, it is most desirable for the techniques to be “scalable”, i.e. applicable in conjunction with large-scale manufacturing devices and procedures.
(81) By way of illustration, the AAVs can be loaded on a positively charged anion-exchange column, such as an N-charged amino or imino resin (e.g. POROS or any DEAE, TMAE, tertiary or quaternary amine, or PEI-based resin) or a negatively charged cation-exchange column (such as HS, SP, CM or any sulfo-, phospho- or carboxy-based cationic resin). The column can be washed with a buffer. The column can be eluted with a gradient of increasing NaCl concentration or a gradient of decreasing pH and fractions can be collected and assayed for the presence of AAV and/or contaminants.
(82) Other procedures can be used in place of or, preferably, in addition to the above-described

anion and cation exchange procedures, based on inter-molecular associations mediated by features other than charge as is known in the art. Such other procedures include intermolecular associations based on ligand-receptor pairs (such as antibody-antigen or lectin-carbohydrate interactions), as well as separations based on other attributes of the molecules, such as molecular sieving chromatography based on size and/or shape.

(83) The pool of AAV-containing fractions eluted from a column as described above can be concentrated and purified by tangential flow filtration (TFF). The preparation is filtered through a membrane, and the product is retained. The retained material can be diafiltered using the membrane with successive washes of a suitable buffer. The final sample is highly enriched for the product and can be filtered and stored for use.

(84) Transfection with the vector plasmids can occur on the day cell seeding is performed, or may be done on day one, day two, day three, day four, or day five post cell seeding depending on the cell seeding density.

(85) In one embodiment of the invention, the host cells of choice are triple-transfected with at least one vector encoding a heterologous nucleotide sequence of interest, at least one vector encoding AAV rep and cap genes, and at least one vector encoding adenoviral accessory functions. In a particular embodiment, triple-transfection is performed using PEI transfection.

(86) Packaging of plasmid backbone sequences (reverse packaging) during rAAV production by transient transfection is a common problem well known in the art. The term “reverse packaging” as used herein refers to the encapsidation of the backbone of the ITR containing plasmid instead of the heterologous sequence. Indeed, for AAV vectors, the predominant species of packaged DNA impurities are the plasmid sequences adjacent to the ITR flanking the expression cassette, likely generated by reverse packaging from the ITR. The presence of these particles in the rAAV stocks may represent a safety issue for clinical applications.

(87) As shown in Example 1 below, the combination of optimized plasmids for the transfection, results in higher yield than production with the standard plasmids commonly used for AAV vector production and lower reverse packaging of bacterial sequences. In particular, the inventors have found that when cells are transfected with three plasmid vectors wherein a first plasmid vector i) comprises an heterologous nucleotide sequence flanked by ITRs and a stuffer DNA sequence located outside said ITRs, wherein said stuffer sequence has a length between 4400 Kb and 4800 Kb so that the plasmid backbone size is above 5 Kb; a second plasmid vector ii) comprising from 5′ to 3′ an AAV rep coding region, an AAV cap coding region and a nucleotide sequence comprising a AAV p5 promoter region, and a third plasmid vector iii) comprising adenovirus helper functions including VA-RNA, E2A y E4 sequences, wherein said plasmid does not contain E3, pTB(E2B), and Ad ITR, protease sequences, rAAV production is significantly increased. In this regard, according to the results shown below, AAV yield is augmented by three times when the combination of these three optimized plasmids is used instead of a combination of standard plasmids. Moreover, reverse packaging is also significantly reduced when compared with a triple transfection using standard plasmids well-known for the skilled person in the art, i.e. a non-oversized plasmid comprising the heterologous nucleotide sequence, a pRepCap vector, for example a pRep2Cap9 vector, and a plasmid containing adenoviral sequences required for AAV production (VA-RNA, E2A y E4). As mentioned before, the optimized helper plasmid used in the present invention contains the adenovirus helper functions including VA-RNA, E2A y E4 sequences required for AAV production. Said plasmid has been modified so that it does not contain certain sequences including E3, pTB(E2B), and Ad ITR sequences.

(88) As it is clearly shown in FIG. **3**, the use of an optimized plasmid containing an heterologous nucleotide sequence, wherein said plasmid is an oversized plasmid containing a stuffer DNA sequence, does not improve vector genome yield when used in combination with non-optimized plasmids i.e. with the standard vectors used for triple transfection. Thus, the use of an optimized oversized plasmid alone does not result in improvement of vector genome yield by triple

transfection.
(89) However, the combination of this optimized oversized plasmid with an optimized plasmid containing adenovirus helper functions (pAdhelper861) or an optimized plasmid containing AAV rep coding region and AAV cap coding region (pRepCap9-809) improved vector genome yield (see FIG. **3**). Furthermore, when the three optimized vectors are combined, vector genome yield is greatly enhanced (see FIG. **3**).
(90) Moreover, the inventors have surprisingly found a reduction of reverse packaging of bacterial sequences when co-transfection of the optimized oversized plasmid (pcohSgsh-900) with either one or both optimized helper plasmids, pRepCap9-809 and/or pAdhelper861 is performed (FIG. **4**). In particular, the use of the triple combination of these optimized plasmids results in a greater reduction in reverse packaging (FIGS. **4** and **5**). Indeed, the reduction in the percentage of bacterial sequences due to the use of optimized helper plasmids (pAdhelper861) and/or (pRepCap9-809) was more pronounced when the optimized plasmid containing an oversized backbone (pcohSgsh-900) was co-transfected (FIG. **5**).
(91) The term "oversized backbone" as used herein refers to a plasmid backbone with a size in base pairs above the limit of encapsidation of the AAV vectors (approx. 4700 bp).
(92) In a preferred embodiment, said second plasmid vector used in the method of the invention comprises an AAV Rep2 and an AAV Cap9 coding regions. In a preferred embodiment, said plasmid vector is pRepCap-809 with accession number DSM 32 as set forth in SEQ ID NO: 5.
(93) Thus, in a particular embodiment of the method of the invention, in step a) the following three-plasmid vectors are used: i) a first plasmid vector comprising a heterologous nucleotide sequence flanked by ITRs and a stuffer DNA sequence located outside said ITRs, wherein said stuffer sequence has a length between 4400 Kb and 4800 Kb so that the plasmid backbone size is above 5 Kb; ii) a second plasmid vector comprising from 5′ to 3′ an AAV rep coding region, an AAV cap coding region and a nucleotide sequence comprising an AAV p5 promoter region; and iii) a third plasmid vector comprising adenovirus helper functions including VA-RNA. E2A y E4 sequences, wherein said plasmid does not contain E3, pTB(E2B), and Ad ITR protease sequences.
(94) In a particular embodiment, the stuffer DNA sequence is located adjacent to one of the ITRs in the first plasmid vector composing the heterologous nucleotide sequence.
(95) In another particular embodiment, the plasmid backbone size is between 7000 bp and 7500 bp. In a more particular embodiment, the plasmid backbone size is between 7000 bp and 7200 bp.
(96) In a preferred embodiment, said first plasmid vector does not contain an F1Ori nucleotide sequence in the backbone sequence.
(97) In a more preferred embodiment, said first plasmid vector is pcohSgsh-900 with accession number DSM 32967, as set forth in SEQ ID NO: 2.
(98) According to another embodiment of the method of the invention, it is possible to re-transfect the remaining cells in the cell culture with two plasmid vectors so that the production of AAVs can be prolonged as explained above.
(99) Thus, the invention also refers to a method according to the invention wherein in step a) the following two plasmid vectors are used: i) a first plasmid vector comprising a heterologous nucleotide sequence flanked by ITRs and a stuffer DNA sequence located outside said ITRs, preferably, adjacent to one ITR, wherein said stuffer sequence has a length between 4400 Kb and 4800 Kb so that the plasmid backbone size is above 5 Kb, preferably between 7000 bp and 7500 bp, more preferably between 7000 bp and 7200 bp; and ii) a second plasmid vector comprising an AAV rep coding region, an AAV cap coding region, a nucleotide sequence comprising an AAV p5 promoter region, and adenovirus helper functions including VA-RNA, E2A and E4 sequences, wherein said plasmid does not contain E3, pTB(E2B), and Ad ITR Protease sequences.
(100) In another particular embodiment, said second plasmid vector comprises from 5′ to 3′ the adenovirus helper functions including VA-RNA, E2A and E4 sequences, an AAV rep coding region, an AAV cap coding region, a nucleotide sequence comprising an AAV p5 promoter region.

(101) In a particular embodiment, said first plasmid vector does not contain an F1Ori nucleotide sequence in the backbone sequence.
(102) In another particular embodiment, said second plasmid vector comprises AAV Rep2 and AAV Cap9 coding regions.
(103) As mentioned before, the combination of optimized plasmids for the transfection, results in higher yield than production with the standard plasmids commonly used for AAV vector production and lower reverse packaging of bacterial sequences.
(104) In the Examples below it is shown that when production of rAAVs is performed using i) a plasmid vector comprising the heterologous nucleotide sequence flanked by ITRs; ii) a plasmid vector comprising from 5′ to 3′ an AAV rep coding region, an AAV cap coding region and a nucleotide sequence comprising a AAV p5 promoter region; and iii) a plasmid vector comprising adenovirus helper functions including VA-RNA, E2A y E4 sequences, wherein said plasmid does not contain E3, pTB(E2B), and Ad ITR protease sequences, reverse packaging is significantly reduced and moreover, productivity is increased, when compared with standard plasmids used for the production of AAVs by triple transfection.
(105) Thus, in another aspect the invention refers to a method for the production of a recombinant AAV said method comprising the steps of: a) co-transfecting a suitable cell with i) a first plasmid vector comprising a heterologous nucleotide sequence flanked by ITRs; ii) a second plasmid vector composing from 5′ to 3′ an AAV rep coding region, an AAV cap coding region and a nucleotide sequence comprising an AAV p5 promoter region: and iii) a third plasmid vector comprising adenovirus helper functions including VA-RNA, E2A y E4 sequences, wherein said plasmid does not contain E3, pTB(E2B), and Ad 1TR protease sequences; b) culturing said cell under conditions allowing AAV replication and packaging; and c) recovering the AAVs produced in step b).
(106) In a preferred embodiment, said first plasmid vector i) is characterized in that the plasmid backbone size is above 5000 pb, preferably between 7000 bp and 7500 bp, more preferably, between 7000 bp and 7200 bp, and that it comprises a stuffer DNA sequence located outside said ITRs, preferably adjacent to one of the ITRs, and wherein said stuffer sequence has a length between 4400 pb and 4800. In a more preferred embodiment, said first plasmid vector i) does not contain an F1Ori nucleotide sequence in the backbone sequence. In a more preferred embodiment of the invention, said first plasmid is pcohSgsh-900 with accession number DSM 32967, as set forth in SEQ ID NO: 2.
(107) In another preferred embodiment, said second plasmid vector ii) comprises AAV Rep2 and AAV Cap9 coding regions. In a preferred embodiment, said plasmid vector is pRepCap-809 with accession number DSM 32969 as set forth in SEQ ID NO: 5.
(108) In another preferred embodiment, said third plasmid vector iii) is pAdHelper861 with accession number DSM 32965 as set forth in SEQ ID NO: 6.
(109) As mentioned above, in another particular embodiment of the invention, step b) is performed culturing said cell in suspension in agitated liquid medium.
(110) In another aspect, the invention also refers to a plasmid vector comprising: a) a heterologous nucleotide sequence flanked by ITRs; and b) a stuffer DNA sequence located outside said ITRs and adjacent to one ITR, wherein said stuffer sequence has a length between 4400 Kb and 4800 Kb, more preferably, between 4500 Kb and 4700 Kb, even more preferably between 4600 Kb and 4700 Kb, so that the plasmid backbone size is above 5 Kb, preferably between 7000 bp and 7500 bp; more preferably, between 7000 bp and 7200 bp;
wherein said plasmid vector does not contain a F1Ori nucleotide sequence in the backbone sequence. #n a more preferred embodiment of the invention, said plasmid is pcohSgsh-900 with accession number DSM 32967, as set forth in SEQ ID NO: 2.
(111) In another particular aspect, the invention refers to a helper plasmid vector containing the adenoviral sequences E2, E4 and VA-RNA wherein said plasmid vector does not contain E3, pTB(E2B), and Ad ITR Protease sequences, More particularly, this plasmid, contains a region of

731 bp that includes the VA genes, a 5346 bp region that includes the E2A gene and the E4 genes and their regulatory regions are included into a 3181 bp fragment. The plasmid backbone contains a high copy number bacterial origin of replication. In a more preferred embodiment, the invention refers to the plasmid vector of SEQ ID NO: 6 which corresponds to pAdHelper861 deposited under accession number DSM 32965.

(112) Use of rAAV for Gene Therapy

(113) The rAAV viral vectors of this invention can be used for administration to an individual for purposes of gene therapy. Suitable diseases for gene therapy include but are not limited to those induced by viral, bacterial, or parasitic infections, various malignancies and hyperproliferative conditions, autoimmune conditions, and congenital deficiencies. In a preferred embodiment, the rAAV vectors produced according to the method of the invention are used for the treatment of lysosomal storage diseases (LSDs). More preferably, said vectors are helpful for the treatment of mucopolysaccharidoses (MPS). In a preferred embodiment, said heterologous nucleotide sequence is a sequence that codifies for an enzyme useful for the treatment of MPS. In particular, said sequence is selected from the group consisting of human α-L-iduronidase, human heparan sulfamidase, human N-sulfoglucosamine sulfohydrolase, human N-acetylglucosaminidase, alpha, human heparan-o-glucosaminide, human N-acetyltransferase, human Iduronate 2-sulfatase (IDS), human N-acetylglucosamine 6-sulfatase, human galactose-6-sulfate sulfatase, human β-galactosidase, human N-acetylgalactosamine-4-sulfatase, β-glucuronidase, and human hyaluronidase.

(114) Gene therapy can be conducted to enhance the level of expression of a particular protein either within or secreted by the cell. Vectors of this invention may be used to genetically alter cells either for gene marking, replacement of a missing or defective gene, or insertion of a therapeutic gene. Alternatively, a polynucleotide may be provided to the cell that decreases the level of expression. This may be used for the suppression of an undesirable phenotype, such as the product of a gene amplified or overexpressed during the course of a malignancy, or a gene introduced or overexpressed during the course of a microbial infection. Expression levels may be decreased by supplying a therapeutic polynucleotide comprising a sequence capable, for example, of forming a stable hybrid with either the target gene or RNA transcript (antisense therapy), capable of acting as a ribozyme to cleave the relevant mRNA or capable of acting as a decoy for a product of the target gene.

(115) Pharmaceutical compositions can be supplied as liquid solutions or suspensions, as emulsions, or as solid forms suitable for dissolution or suspension in liquid prior to use. The carrier is, for instance, water, or a buffered saline solution, with or without a preservative.

(116) The pharmaceutical compositions may be lyophilized for re-suspension at the time of administration or in solution. In a preferred embodiment, the pharmaceutical composition of the invention is a suspension for injection. The final product is then formulated in a suitable buffer, filled in vials and stored until use.

(117) Although not required, pharmaceutical compositions may optionally be supplied in unit dosage form suitable for administration of a precise amount.

(118) Having described the invention in general terms, it will be more easily understood by reference to the following examples which are presented as an illustration and are not intended to limit the present invention in any way.

EXAMPLES

(119) The practice of the present invention will employ, unless otherwise indicated, conventional techniques of molecular biology, virology, animal cell culture and biochemistry which are within the skill of the art. Such techniques are explained fully in the literature.

Example 1

Triple Transfection Using Standard Plasmid Vectors vs Optimized Vectors

(120) 1.1. Plasmid Description

(121) Plasmid pcohSgsh-827 (SEQ ID NO: 1)
(122) Plasmid containing a 1512 bp codon optimized cDNA that codifies for the human N-sulfoglucosamine sulfohydrolase under the control of the 1756 bp CAG promoter and a 519 bp fragment corresponding to the rabbit b-globin pA. These three elements are flanked by the AAV 2 ITR sequences. The plasmid backbone size is 3156 bp (standard backbone size) and comprises a 966 bp ampicillin resistance gene, a 514 bp bacteriophage origin of replication and a 589 bp high copy number bacterial origin of replication.
(123) Plasmid pcohSgsh-900 (SEQ ID NO: 2)
(124) Oversized plasmid containing a 1512 bp codon optimized cDNA that codifies for the human N-sulfoglucosamine sulfohydrolase under the control of the 1756 bp CAG promoter and a 519 bp fragment corresponding to the rabbit b-globin pA. These three elements are flanked by the AAV 2 ITR sequences. The plasmid backbone is 7073 bp, comprises the 1043 bp kanamycin resistance gene, a 589 bp high copy number bacterial origin of replication and additionally includes a 4657 bp random sequence.
(125) Plasmid pohIDS-874 (SEQ ID NO: 3)
(126) Oversized plasmid containing a 1653 bp codon optimized cDNA that codifies for the human Iduronate 2-sulfatase under the control of the 1756 bp CAG promoter and a 519 by fragment corresponding to the rabbit b-globin pA. These three elements are flanked by the AAV 2 ITR sequences. The plasmid backbone is 7073 bp and comprises the 1043 bp kanamycin resistance gene, a 589 bp high copy number bacterial origin of replication and additionally includes a 4657 bp random sequence.
(127) Plasmid pRepCap9-808 (SEQ ID NO: 4)
(128) Plasmid encoding the rep2 and the cap9 proteins with the P5 promoter in its original position. This plasmid includes a 1866 bp fragment corresponding to the AAV2 Rep gene followed by a sequence of 2211 bp encoding the AAV9 Cap gene. The 130 bp AAV2 P5 promoter is located on its original position upstream of the Rep gene. The plasmid backbone comprises the ampicillin resistance gene and a high copy number bacterial origin of replication.
(129) Plasmid pRepCap9-809 (SEQ ID NO: 5)
(130) Plasmid encoding the rep2 and the cap9 proteins with the P5 promoter after the Cap9 gene. This plasmid includes a 1866 bp fragment corresponding to the AAV2 Rep gene followed by a sequence of 2211 bp encoding the AAV9 Cap gene. A sequence of 130 bp corresponding to the AAV2 P5 promoter is located downstream of the Cap gene. The plasmid backbone comprises the kanamycin resistance gene and a high copy number bacterial origin of replication.
(131) Plasmid pXX6
(132) Plasmid containing the adenoviral sequences (E2, E4 and VA-RNA) which are required for the AAV production. The plasmid backbone comprises the ampicillin resistance gene and a high copy number bacterial origin of replication (Xiao et al. J Virol. 1998 March; 72(3): 2224-2232). Plasmid pXX6-80 derives from pXX6 where only the backbone was changed. pXX6-80 map and sequence are well described in the literature.
(133) Plasmid pAdHelper861 (SEQ ID NO: 6)
(134) Plasmid containing the adenoviral sequences (E2, E4 and VA-RNA) which are required for the AAV production, it contains a region of 731 bp that includes the VA genes, a 5346 bp region that includes the E2A gene and the E4 genes and their regulatory regions are included into a 3181 bp fragment. The plasmid backbone comprises the kanamycin resistance gene and a high copy number bacterial origin of replication.
(135) 1.2. AAV9-CAG-cohSgsh Vector Production by Triple Transfection
Materials and Methods
(136) AAV9-CAG-cohSgsh vectors were produced by transient triple-transfection using PEI-MAX as transfection agent as previously described (Ayuso E, et at. Gene Ther, 2010 April; 17(4):503-10). Briefly, HEK293 adherent cells at confluency were transfected with equimolar quantities of three

plasmids: pcohSgsh-827, pRepCap9-808 and pXX6 (standard plasmids) or pcohSgsh-900, pRepCap9-809 and pAdHelper861 (optimized plasmids). PEE-MAX was used at a PEI:DNA ratio of 2:1. Cells were harvested 72 hours post-transfection and lysed by three cycles of freeze and thaw to release rAAV vectors from inside the cells (see Example 2 below). Production of rAAV vectors using suspension HEK293 cells has also been described (Joshua C Grieger, et al. Mol Ther. 2016 February; 24(2):1287-297),

(137) Vector genomes were quantified by Taqman qPCR as previously described (Ayuso et al. cited supra) using primers and probe specific for the rabbit beta-globin poly A sequence present in the expression cassette. Also, reverse packaging was quantified by Taqman qPCR with primers and probe specific for the antibiotic resistant promoter sequence present in all the plasmid backbones used.

(138) Statistical analyses were performed with GraphPad Prism 7.00.

(139) TABLE-US-00001 TABLE 1 List of plasmids used for the triple-transfection. Name Plasmid type pcohSgsh-827 Standard pRepCap9-808 Standard pXX6 Standard pcohSgsh-900 Optimized pRepCap9-809 Optimaed pAdHelper861 Optimized

Results

(140) Production of recombinant AAV9 vectors AAV9-CAG-cohSgsh by triple transfection using optimized plasmids, as described above, results in higher yield than production with the standard plasmids commonly used for AAV vector production (FIG. **1**).

(141) AAV9-CAG-cohSgsh vectors produced by triple transfection with the optimized plasmids result in lower reverse packaging of bacterial sequences than those obtained with the standard plasmids commonly used for AAV production (FIG. **2**).

(142) The use of an optimized plasmid containing an heterologous nucleotide sequence, wherein said plasmid is an oversized plasmid containing a stuffer DNA sequence (pcohSgsh-900), does not improve vector genome yield when used in triple transfection in combination with non-optimized plasmids. i.e. with the standard vectors used for triple transfections. Thus, the use of an optimized oversized plasmid alone does not result in improvement of vector genome yield by triple transfection (FIG. **3**).

(143) However, the combination of this optimized oversized plasmid with an optimized plasmid containing adenovirus helper functions (pAdhelper861) or an optimized plasmid containing AAV rep coding region and AAV cap coding region (pRepCap9-809) improved vector genome yield (see FIG. **3**). Furthermore, when the three optimized vectors are combined, vector genome yield is greatly enhanced (see FIG. **3**).

(144) Moreover, the inventors have surprisingly found a reduction of reverse packaging of bacterial sequences when co-transfection of the optimized oversized plasmid (pcohSgsh-900) with either one or both optimized helper plasmids, pRepCap9-809 and/or pAdhelper861 is performed (FIG. **4**). In particular, the use of the triple combination of these optimized plasmids results in a greater reduction in reverse packaging (FIGS. **4** and **5**) Indeed, the reduction in the percentage of bacterial sequences due to the use of optimized helper plasmids (pAdhelper861) and/or (pRepCap9-809) was more pronounced when the optimized plasmid containing an oversized backbone (pcohSgsh-900) was co-transfected (FIG. **5**).

Example 2

Triple Transfection and Extended Gene Expression (EGE)

(145) 2.1. Cell Line, Media, and Culture Conditions

(146) The cell line used is a serum-free suspension-adapted HEK 293 cell line (HEK 293SF-3F6) provided by the Biotechnology Research Institute of National Research Council of Canada (Montreal, Canada). Cells are cultured in Freestyle F17 medium (Invitrogen, Carlsbad, CA) supplemented with, Glutamax 8 mM, 0.1% Pluronic1 (Invitrogen) and 0.05 ng/L IGF. Cells are routinely maintained in 125-mL disposable polycarbonate erlenmeyer flasks (Corning, Steuben, NY) in 20 mL of culture medium. Flasks are shaken at 130 rpm using an orbital shaker (Kuhner

shakers, Switzerland) placed in an incubator maintained at 37° C. in a humidified atmosphere of 5% CO.sub.2 in air. Cell count and viability are determined using Nucleocounter NC-3000 (Chemometec, Denmark).
(147) 2.2. Transient Transfection (TGE)
(148) HEK 293 suspension cells are transiently transfected using PEIPro (Polysciences, Warrington, PA). HEK 293 cells are seeded at 0.5.Math.10.sup.6 cells/mL in 125-mL disposable flasks, grown to 2.Math.10.sup.6 cells/mL and transfected with 0.76 μg of pAdHelper861/mL of culture, 0.77 μg of pRepCap9-809/mL of culture and 0.38 μg of pohIDS-874/mL or 0.38 μg of pcohSgsh-900/mL of culture and a DNA to PEI mass ratio of 1.2. PEI/DNA complexes are formed by rapidly adding PEI to DNA, both diluted in fresh culture media to attain the same volume (complex mixture volume is 5% of the total volume of the culture to be transfected). The mixture is incubated for 15 min at room temperature to allow complex formation poor to its addition to the cell culture.
(149) 2.3. Extended Gene Expression Methodology for coh-IDS Gene
(150) The Extended Gene Expression (EGE) production strategy consists in performing repeated rounds of transfection to achieve a sustained level of gene expression over time as opposed to the conventional TGE approach which entails a single transfection round. After the first transfection, using the same plasmids and conditions as described above in 2.2, retransfection rounds were performed every 48 hours using the same plasmid and PEI concentrations after a complete medium exchange performed by centrifugation (at 300×g during 5 minutes). By doing these medium exchanges, AAVs are secreted to the supernatant of the culture and can be harvested every time the medium exchange before each retransfection is performed.
(151) The results show (See FIG. **3**) a 3-fold increase in total vg harvested from the supernatant when EGE methodology is performed. Total vg recovered after Batch (single transfection) was 8E11 vg obtained after cell lysis. Total vg recovered after EGE strategy was 2.5E12 vg from the supernatant, this being an advantage from the purification point of view as cells do not need to be lysed to harvest the AAVs and thus less contamination with host cell DNA and host cell protein is expected.
(152) 2.4. Extended Gene Expression Methodology for cob-Sgsh Gene
(153) The EGE methodology has also been tested for the gene coh-Sgsh as explained above in 2.3.
(154) The results show (See FIG. **4**) a 1.7-fold increase in total vg harvested from the supernatant when EGE methodology is performed. Total vg recovered after Batch (single transfection) was 1.9E12 vg obtained after cell lysis. Total vg recovered after EGE strategy was 3.18E12 vg from the supernatant. AAV production followed similar patterns that for phIDS (see 2.3 above) thus, supporting the general applicability of the EGE strategy.

Claims

1. A method for the production of a recombinant Adeno-Associated Virus (AAV) vector, the method comprising the steps of: a) co-transfecting a suitable cell culture with three plasmid vectors: i) a first plasmid vector characterized in that the plasmid backbone size is above 5000 bp, and comprising a heterologous nucleotide sequence flanked by inverted terminal repeats (ITRs) and a stuffer DNA sequence located outside said ITRs, wherein said stuffer sequence has a length between 4400 bp and 4800 bp, wherein said plasmid vector does not contain an F1Ori nucleotide sequence in the backbone sequence, and wherein said plasmid vector is pcohSqsh-900 with accession number DSM 32967 having the sequence as set forth in SEQ ID NO: 2; ii) a second plasmid vector comprising from 5′ to 3′ an AAV rep coding region, an AAV cap coding region and a nucleotide sequence comprising a AAV p5 promoter region; and iii) a third plasmid vector comprising adenovirus helper functions including VA-RNA, E2A and E4 sequences, wherein said plasmid does not contain E3, pTB (E2B), and Ad ITR and protease sequences, and wherein said

plasmid vector is pAdHelper-861 having accession number DSM 32965 having the sequence as set forth in SEQ ID NO: 6; b) culturing said cells under conditions allowing AAV replication and packaging; c) recovering recombinant AAVs produced in step b) and retaining the cells in the cell culture under conditions allowing further division and growth; d) re-transfecting the cells according to step c) with the plasmid vectors according to step a); and e) repeating steps b) to c).

2. The method of claim 1, wherein in step b) recombinant AAVs are secreted to the supernatant of the cell culture.

3. The method of claim 1, wherein the cell media of the cell culture is exchanged before step d).

4. The method of claim 3, wherein cell media exchange is performed by perfusion.

5. The method of claim 1, wherein steps d), b) and c) are repeated at least one more time after recovering step c).

6. The method of claim 1, wherein step b) is performed culturing said cell in suspension in agitated liquid medium.

7. A method for the production of a recombinant AAV vector the method comprising the steps of: a) co-transfecting a suitable cell with i) a first plasmid vector characterized in that the plasmid backbone size is above 5000 bp, and comprising a heterologous nucleotide sequence flanked by ITRs and a stuffer DNA sequence located outside said ITRs, wherein said stuffer sequence has a length between 4400 bp and 4800 bp, wherein said plasmid vector does not contain an F1Ori nucleotide sequence in the backbone sequence, and wherein said plasmid vector is pcohSgsh-900 with accession number DSM 32967 having the sequence as set forth in SEQ ID NO: 2; ii) a second plasmid vector comprising from 5′ to 3′ an AAV rep coding region, an AAV cap coding region and a nucleotide sequence comprising an AAV p5 promoter region; and iii) a third plasmid vector comprising adenovirus helper functions including VA-RNA, E2A and E4 sequences, wherein said plasmid does not contain E3, pTB (E2B), and Ad ITR and protease sequences, and wherein said plasmid vector is pAdHelper-861 having accession number DSM 32965 having the sequence as set forth in SEQ ID NO: 6; b) culturing said cell under conditions allowing AAV replication and packaging; and c) recovering recombinant AAVs produced in step b).

8. The method of claim 7, wherein said second plasmid vector (ii) comprises AAV Rep2 and AAV Cap9 coding regions.

9. A plasmid vector comprising: a) a heterologous nucleotide sequence flanked by inverted terminal repeats (ITRs); and b) a stuffer DNA sequence located outside said ITRs and adjacent to one ITR, wherein said stuffer sequence has a length between 4400 bp and 4800 bp so that the plasmid backbone size is above 5 Kb; wherein said plasmid vector does not contain an F1Ori nucleotide sequence in the backbone sequence, and wherein said plasmid vector is pcohSgsh-900 with accession number DSM 32967 having the sequence as set forth in SEQ ID NO: 2.

10. A plasmid vector comprising adenovirus helper function sequences selected from the group consisting of VA-RNA, E2A and E4 sequences, wherein said plasmid does not contain E3, pTB (E2B), and Ad ITR and protease sequences, and wherein said plasmid vector is pAdHelper-861 having accession number DSM 32965 and having the sequence as set forth in SEQ ID NO: 6.